

ANEKANT EDUCATION SOCIETY'S  
**TULJARAM CHATURCHAND COLLEGE OF ARTS, SCIENCE AND  
COMMERCE, BARAMATI**  
**(EMPOWERED AUTONOMOUS)**



**A RESEARCH PROJECT  
REPORT ON**

**“Determinants of Work–Life Balance and its Relationship with Job Satisfac-  
tion among Working Adults”**

Submitted to  
Department of statistics

by

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Year 2025-2026

Anekant Education Society's  
**Tuljaram Chaturchand College Arts, Science & Commerce Baramati**  
**(Empowered Autonomous)**



**DEPARTMENT OF STATISTICS**

**CERTIFICATE**

This is to certify that the research project titled **“Determinants of Work–Life Balance and its Relationship with Job Satisfaction among Working Adults”**

Submitted by..... for the  
award of the degree of Masters of Science in Statistics in partial fulfilment of Research Project examination conducted by Department of Statistics Tuljaram Chaturchand College, Baramati was carried out by the candidate under the supervision of Dr. Nilambari A. Jagtap, during the academic year 2025-2026.

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## **ABSTRACT**

This study focuses on understanding the factors that affect Work-Life Balance (WLB) and how it is related to job satisfaction among working adults across different sectors. In today's fast-changing work environment, maintaining a balance between professional and personal life has become difficult due to increased workload, use of technology, and changing work patterns.

The main aim of this study is to identify the key factors influencing work-life balance and to examine how it differs across various demographic and occupational groups. The data was collected using a structured questionnaire through both online (Google Forms) and offline survey methods, which helped in getting a diverse set of responses.

Exploratory Data Analysis (EDA) was carried out to understand the data and identify patterns. Statistical methods such as correlation analysis, non-parametric tests, and factor analysis were used to study relationships and underlying factors. The factor analysis revealed two main dimensions of work-life balance, which explain a significant portion of the data.

The findings show that factors like family time, workload, and workplace support have a strong impact on work-life balance. Differences were observed across occupational groups, while gender differences were minimal. A positive relationship between work-life balance and job satisfaction was also identified.

Overall, the study concludes that maintaining a good work-life balance depends on both manageable work conditions and supportive environments. Improving these factors can lead to better well-being, higher job satisfaction, and improved productivity.

### **Keywords: -**

Work-Life Balance, Job Satisfaction, Work Pressure, Personal Time, Workplace Support, Factor Analysis, Machine Learning, Recommendation System

## **MOTIVATION**

In today's fast-changing world, maintaining a healthy work-life balance has become very important. Due to increasing job pressure, longer working hours, and changing lifestyles, many people find it difficult to manage their work along with their personal and social life. This often leads to stress, reduced productivity, and lower overall well-being.

With growing competition and modern work culture, the gap between work and personal life is slowly reducing. People are often expected to stay connected to work even after working hours, which makes it harder to maintain a proper balance. This imbalance not only affects individuals but also impacts job satisfaction and overall performance.

The main motivation behind this study is to understand what factors affect work-life balance in a proper and structured way using data. By analysing real responses, this study focuses on important aspects like workload, stress, personal time, and workplace support.

This research also aims to understand how these factors are connected and how they influence a person's overall well-being. The findings can help organizations and individuals take better decisions to improve work-life balance.

Overall, the study is motivated by the need to promote a healthier and more balanced lifestyle, which can improve both personal well-being and work efficiency.

## **INTRODUCTION**

In today's fast-changing and highly competitive world, the concept of work-life balance has become more important than ever before. Work-life balance refers to the ability of an individual to manage their professional responsibilities along with their personal life in a healthy and satisfying way. It is not just about dividing time between work and personal life, but also about maintaining harmony between the two so that neither aspect negatively affects the other. With increasing job demands, longer working hours and the constant use of technology, maintaining this balance has become a major challenge for many individuals.

The importance of work-life balance has grown significantly due to changes in work culture and lifestyle. Many organizations expect employees to be available beyond traditional working hours, which often leads to stress and burnout. At the same time, individuals also have personal responsibilities such as family, health and social commitments. When there is an imbalance between work and personal life, it can lead to physical and mental health issues, reduced job satisfaction and lower overall quality of life. Therefore, achieving a proper balance is essential not only for employees but also for organizations that aim to improve productivity and employee retention.

Work-life balance plays a crucial role in influencing an employee's mental health, productivity and overall well-being. When employees are able to manage their work and personal life effectively, they tend to experience lower levels of stress and anxiety. This, in turn, improves their focus, efficiency and performance at work. On the other hand, poor work-life balance can result in fatigue, emotional exhaustion and decreased motivation. Employees who are constantly overworked may struggle to maintain concentration and may become less productive over time. Thus, a healthy balance between work and personal life contributes to both individual well-being and organizational success.

Another important aspect to consider is that work-life balance is not experienced in the same way across all sectors. Different types of jobs come with different levels of workload, flexibility and job security. For example, employees working in the IT sector often face long working hours and tight deadlines, which can make it difficult to maintain a balance. Similarly, private sector employees may experience pressure to meet targets and perform consistently. In contrast, government jobs are often considered more stable with fixed working hours, which may provide better work-life balance. Daily wage workers and individuals involved in farming may

face different challenges, such as income uncertainty and physically demanding work, which affect their personal lives in unique ways.

Self-employed individuals also experience work-life balance differently, as they have greater control over their work but may face financial risks and irregular schedules. In sectors like healthcare and emotional involvement, which can impact their personal well-being. Similarly, employees in banking and finance may face high pressure due to deadlines and financial responsibilities. Therefore, it is important to study work-life balance across sectors to understand these differences and identify common challenges individuals face.

In the Indian context, the study of work-life balance has become increasingly relevant. With rapid economic growth, urbanization and changing work patterns, employees in India are facing new kinds of pressures and expectations. The rise of digital technology has blurred the boundaries between work and personal life, making it difficult for individuals to disconnect from work. Additionally, cultural factors and family responsibilities also play a significant role in shaping how individuals manage their work and personal life. As a result, there is a growing need to study work-life balance in India to develop strategies that can improve the quality of life of employees across different sectors.

The present study aims to examine work-life balance among individuals working in various sectors, including IT, private non-IT, government, daily wage work, farming, self-employment, healthcare, education, banking and finance. To ensure a comprehensive understanding, data for this study were collected using both online and offline methods. A structured questionnaire was designed and distributed through Google Forms to collect responses from participants across different locations. In addition to this, offline field surveys were conducted to gather data from individuals who may not have access to online platforms. This approach helped in obtaining a diverse and representative sample from multiple sectors.

Overall, this study focuses on understanding the factors affecting work-life balance and how it varies across different types of occupations. By analysing the collected data, the study aims to provide meaningful insights into the challenges faced by individuals and suggest possible ways to improve work-life balance. This research is expected to be useful for both employees and organizations in creating a healthier and more productive work environment.

Furthermore, maintaining work-life balance has become increasingly important in ensuring long-term career sustainability, mental stability and overall life satisfaction. In the absence of a proper balance, individuals may experience prolonged stress, which can gradually lead to



burnout, decreased productivity and serious health issues such as anxiety, depression and fatigue. In recent years, organizations have started recognizing the importance of employees' well-being and are adopting various measures such as flexible working hours, remote work options, wellness programs and supportive work environments to help employees manage their responsibilities more effectively. However, achieving work-life balance is not solely the responsibility of organizations; individuals must also play an active role by managing their time efficiently, setting clear boundaries between work and personal life and prioritizing their physical and mental health. The role of technology is also significant in this context, as while it has increased convenience and connectivity, it has also made it difficult for employees to disconnect from work. Therefore, it is essential to use technology mindfully to avoid work intrusion into personal time. A well-maintained work-life balance not only improves individual happiness and job satisfaction but also enhances creativity, decision-making ability and overall performance in the workforce and leads to a more sustainable and productive organizational culture.

## **LITERATURE REVIEW**

### **1)Shalini Sathya Prasad & Prof. Sreenivas M. (2019)**

This research paper examines how male and female college teachers perceive and manage their work-life balance in Bengaluru, India. Work-life balance is extremely important for teachers because they not only manage classroom duties but also spend extra hours preparing lessons, checking papers and handling administrative work. Traditionally, women were believed to face more difficulty balancing work and family, but changing social roles have made work-life balance an important concern for men too.

To explore this issue, the researchers collected data from 180 teachers, 90 male and 90 female and used the Work-Life Balance Scale developed by Pareek & Joshi (2010). The study compared scores using an independent sample t-test to identify whether gender plays a significant role in influencing work-life balance.

Interestingly, the results showed that there was no significant difference between men and women in their overall work-life balance index. Even though female teachers scored slightly higher on average, the difference was not statistically meaningful. This suggests that both male and female teachers experience similar levels of balance or imbalance in handling work and family responsibilities in today's changing social environment.

The paper also noted that modern family structures, such as dual-earner households and increasing male involvement in domestic responsibilities, may have reduced the traditional gender gap in work-life expectations. It emphasized that maintaining a positive work-life balance improves job satisfaction, well-being and performance for all employees regardless of gender.

Overall, this research proves that work-life balance is equally important for both men and women and organizations, especially educational institutions, should introduce supportive policies and a healthy working environment to help teachers manage professional and personal demands more effectively.

### **2)Dr. Neeti Sharma (2021)**

This research paper presents a **systematic review of** work-life balance (WLB) studies focusing on gender differences. The main purpose of the study was to identify important variables that influence work-life balance for both men and women and to highlight gaps that future

researchers can address. To achieve this, the author reviewed **92 journal publications** published between **1987 and 2016**, covering research from psychology, management and social domains.

The study revealed **11 major variables** that influence work-life balance, including spouse support, organizational policies, stress, supervisor support, health and well-being, work-family conflict, demographic factors, job satisfaction and flexibility. Findings were that **work-family conflict is more commonly experienced by women**, largely due to traditional gender roles and their higher involvement in caregiving activities. The study also found that organizations with supportive policies, such as flexible work hours, childcare support and employee-centred practices help employees manage work and family demands more effectively.

To compile the review, the author followed a structured search strategy using databases such as EBSCO, Google Scholar, Emerald and ProQuest. Out of **584 initial publications, only 92 met the final inclusion criteria**. The selected studies used various research methods, including quantitative surveys, qualitative interviews and mixed method approaches. The review also noted that most of the research came from India, the United States, Europe and Asian countries, highlighting the global relevance of work-life balance issues.

The findings emphasized that **stress, lack of organizational support, long working hours and conflicting work-family demands** negatively affect both men and women. However, women often experience higher levels of imbalance because of additional responsibilities at home. The paper also identified key gaps in existing research, such as the need for **longitudinal studies**, better sampling methods, more research on **men's experiences**, and studies across diverse industries.

Overall, the research demonstrates that work-life balance is influenced by multiple factors and varies between genders.

### **3)Nikhat Tabasum & Dr. B. S. Hugar (2020)**

This research paper focuses on understanding how men and women differ in managing their work-life balance, especially in the rapidly growing service their work-life balance, especially in the rapidly growing service sector environment of Hyderabad, India. With increasing job opportunities in IT and service industries, working hours have expanded, workloads have intensified and employees often face long and unpredictable schedules. As a result, both men and women experience challenges in balancing professional responsibilities and personal life.

The study used a **questionnaire method** and collected responses from **100 employees** (70 men and 30 women). The objective was to identify whether gender plays a major role in work-life balance issues and which factors can help improve balance. The findings showed that **flexible working hours** were strongly supported by both genders-**97% of men and 90% of women** agreed that flexible timing helps improve harmony between work and personal life.

The study further revealed **that women preferred options like part-time work, childcare facilities, work-from-home and support for home emergencies more than men**. For example, 90% of women believed that child-care facilities are important to maintain work-life balance compared to 76% of men. Women also showed a higher need for support during home emergencies compared to male respondents.

On the other hand, the paper explored how work affects health, sleep and exhaustion. About **53% of men and 30% of women** felt their health was affected due to work pressure and **51% of men and 17% of women** reported feeling exhausted because of work demands. This shows that although women prefer more family-friendly policies, men may be experiencing stronger physical strain from work.

Another important finding was that both men and women acknowledge the importance of a **co-operative work environment**. Around **63% of women and 59% of men** felt that colleagues are supportive and more than half of both genders agreed that they receive cooperation from their supervisors.

Overall, the study concluded that organizations should adopt practices such as **flexible working hours, part-time work opportunities, work from home options, childcare facilities and supportive workplace culture** to enhance work-life balance.

#### **4)Dr. S. Priya (2024)**

This research paper explores the work-life balance (WLB) of women employees across key sectors in Puducherry, including education, healthcare, banking and private organizations. The study was conducted to understand how women manage professional responsibilities alongside domestic and social expectations, given the rising participation of women in the workforce. The author highlights that although Women have made significant progress professionally, they continue to shoulder the majority of household responsibilities, leading to increased strain, role conflict and emotional fatigue.

Using a structured questionnaire, the study gathered data on working hours, job stress, flexibility, family support and organizational practices. The findings revealed that **long working hours, heavy workload and insufficient flexibility** were the primary factors contributing to imbalance among women employees. For many respondents, balancing child care, elderly care and household duties along with professional demands resulted in chronic stress. The pressure was especially high among married women with children, who reported limited time for personal well-being.

The research also emphasized the role of **organizational support systems**, such as flexible working hours, work-from-home options, paid leave, child-care facilities and understanding supervisors. Women who reported supportive supervisors and cooperative colleague had significantly better levels of work-life balance. Family support-especially from spouse and extended family-played another crucial role in reducing burden and improving overall satisfaction.

The study concludes that although women in Puducherry have access to job opportunities across sectors, their work-life balance remains a challenge due to structural, cultural and organizational factors. The author recommends that companies introduce family-friendly policies, employee wellness programs and flexible work arrangements to better support women employees. Improving WLB is not only beneficial for women but also enhances productivity, retention and organizational commitment.

### **5)Ashwin Ram Rajaraman (2018)**

This research investigates work-life balance among IT professionals, a group known for experiencing high stress due to long work hours, tight deadlines and constant technological demands. As the IT sector grows rapidly, employees often struggle to manage personal commitments alongside fast-paced professional environments. The study aims to understand the factors contributing to work-life imbalance and the strategies employees use to cope with job pressure.

The researcher collected data using a questionnaire that assessed working hours, organizational culture, stress levels, family responsibilities, flexibility and satisfaction. The findings showed that **long working hours and a culture of constant availability** were major contributors to imbalance. Many IT employees reported that unpredictable workloads and deadlines interfered with personal life, reduced time with family and resulted in mental and physical fatigue.

Younger professionals, especially those early in their careers, reported greater imbalance due to the pressure to perform and prove themselves.

However, the study also found that organizational policies could significantly improve WLB. Employees who had access **to flexible scheduling, remote work options and supportive managers** demonstrated better balance and reduced stress. The presence of friendly team culture, cooperative colleagues and clear communication channels further enhanced employee satisfaction. The study noted that work-from-home options were particularly helpful for married employees and parents, allowing them to manage personal responsibilities more effectively.

The research concludes that work-life balance among IT professionals is heavily influenced by organizational practices and workplace culture. Employers in the IT sectors must adopt strategies such as flexible work timing, reduced overtime expectations, health and wellness programs and supportive leadership to promote employee well-being. Improving WLB not only benefits employees but also boost productivity, creativity and long-term retention within the organization.

#### **6)Tika Ram & Laxmi**

This article is centred around the validation of a work-life balance scale among Indian female workers. In this study, the effect of certain variables such as age, marital status and family type on work-life balance has been explored. Various statistical tools including Cronbach's alpha, t-test, ANOVA and confirmatory factor analysis have been used for analysing the data.

Results indicate that flexibility at work, family-friendly practices, health promotions activities and work stress are the dimensions of work-life balance. It has been found that younger women have higher work stress than older women and that marital status has an effect on work-life balance but family type has no effect.

#### **7) Hema P.R. (2023)**

This research focuses on the gender disparity in work-life balance in the private sector in India. This paper recognizes the importance how each gender deals with their work and family life.

This study reveals that women find themselves under greater stress because of their roles as caregivers and discrimination at work, whereas men are expected to take precedence in their jobs because they are the breadwinners of the family.

#### **8)Lukasz Marecki (2023)**

The research focuses on how work-life balance affects the productivity and health of employees. It is found that balancing personal and professional aspects of one's life makes workers more satisfied with their work, healthier and committed to the organization they work for.

The findings indicate that work-life balance allows workers to be more productive and motivated at work, avoiding the risk of exhaustion. Flexibility, support from the organization and effective time management are some key factors that help improve productivity.

#### **9)Rida Siddiqui & Seema Rani Saraf (2025)**

The current research paper analyses the link among work-life balance, stress and job performance. It highlights that lack of proper work-life balance results in higher stress, which impacts job performance negatively. The paper evaluates the findings related to international research and theories, such as the Job Demands-Resources Model.

Results reveal that organizational support, flexible working and personality characteristics help to alleviate stress and increase performance. It is concluded that work-life balance serves as a moderator for stress and is important for increasing productivity and job satisfaction.

#### **10)Dr. Dolly Dolai (2015)**

The current research deals with assessing the work-life balance of employees working in the insurance industry of India. Due to high levels of competition and increased work load, the stress level faced by such employees is also quite high. Data were collected from 224 participants through a structured questionnaire with a work-life balance scale of 15 items such as WIPL, PLIW and WPLE.

According to the findings of the study, factors like work hours, workloads and other organizational aspects impact work-life balance adversely. Nonetheless, having flexibility and favourable conditions at work can increase well-being and efficiency. Thus, the findings suggest that appropriate work-life balance contributes to low stress levels and enhanced motivation.

## **OBJECTIVES**

- To construct work-life balance score of individuals.
- To understand the main factors that affect work-life balance such as workload, stress, personal time, and workplace support.
- To compare work-life balance among different groups based on factors like age, gender and occupation.
- To build models that can classify individuals into different work-life balance levels.
- To analyse and predict job satisfaction of individuals using demographic and work-related factors.



## **METHODOLOGY**

We started our project by discussing different topics related to work-life balance and shortlisted a few before finalizing our topic, “**Determinants of Work-Life Balance and its Relationship with Job Satisfaction among Working Adults.**” To gain a deeper understanding of the subject, we studied various research papers related to work-life balance, job satisfaction, and employee well-being. This helped us understand existing research and identify important factors affecting work-life balance.

### **Data Collection**

For this study, primary data was collected using both online and offline methods. A structured questionnaire was designed and distributed through Google Forms, while offline surveys were conducted to include respondents who may not have access to online platforms. This approach ensured a diverse dataset across multiple sectors such as IT, government, private, healthcare, education, banking and finance, farming, and self-employment.

### **Data Organization and Storage**

The collected data was stored and managed in Excel format for easy access and analysis. Data from both Google Forms and offline surveys was compiled into structured Excel sheets containing demographic, work-related, and WLB variables. The dataset was then imported into Python using libraries such as pandas for further processing and analysis. Additionally, proper data backup and organization were maintained by storing raw, processed, and output files in separate folders to ensure data safety and workflow efficiency.

### **Data Cleaning and Preprocessing**

Before analysis, the dataset was cleaned and prepared to improve accuracy and reliability by removing duplicate entries, handling missing values, and converting categorical variables into numerical format for further statistical analysis and model building.

### **Reliability Analysis**

To ensure the consistency of the Work-Life Balance (WLB) scale, Cronbach’s Alpha was calculated, which measures the internal consistency of multiple items. A value greater than 0.7

indicates acceptable reliability, and the results confirmed that the scale used in the study is reliable and suitable for further analysis.

### **Construction of Work-Life Balance (WLB) Score**

A composite WLB index was constructed using multiple Likert-scale items representing different aspects of work and personal life.

#### **Dimensions of WLB**

##### **1. Work Pressure Index (WPI):**

- My workload is manageable during regular working hours
- I often have to work beyond official working hours
- I feel stressed due to my job

##### **2. Personal Life Index (PLI):**

- I get enough rest and sleep on working days
- I am able to balance my work and personal life effectively
- My job allows me enough time for my family/social life

##### **3. Organizational Support Index (OSI):**

- Workplace support

### **Weighted Score Formula**

The final WLB score was calculated as:

$$\text{WLB Score} = 0.40(\text{WPI}) + 0.35(\text{PLI}) + 0.25(\text{OSI})$$

This weighted approach reflects the relative importance of different dimensions.

WLB Score classification into categories: -

- 33rd percentile = 3.2333
- 66th percentile = 3.8833

Classification:

- Low  $\rightarrow \leq 3.2333$
- Moderate  $\rightarrow 3.2333$  to  $3.8833$
- High  $\rightarrow > 3.8833$

This classification helped in better understanding and visualization of WLB levels.

### **Exploratory Data Analysis (EDA)**

Exploratory Data Analysis was carried out to better understand the distribution and relationships within the dataset. The analysis included examining demographic variables such as

gender, occupation, and family type, along with patterns in job satisfaction. The distribution of WLB scores was visualized using density plots, and relationships between variables were explored using correlation heatmaps. This helped in gaining meaningful insights and identifying important patterns in the data.

### **Normality Testing**

Before applying statistical tests, the distribution of the Work-Life Balance (WLB) scores was examined using the Shapiro-Wilk test. This test was used to assess whether the data follows a normal distribution and to guide the selection of appropriate statistical methods for further analysis.

### **Inferential Analysis (Non-Parametric Methods)**

As the Work-Life Balance (WLB) data did not satisfy the normality assumption, non-parametric statistical methods were used. The Mann-Whitney U test was applied to compare WLB scores between two groups, while the Kruskal-Wallis test was used for comparisons across multiple groups. Additionally, post-hoc Dunn's test was performed to identify pairwise group differences.

### **Model Building**

During the model building phase, different machine learning algorithms were trained using the prepared dataset to predict Work-Life Balance (WLB) categories. Models such as Decision Tree (CART), Random Forest, and Ordinal Logistic Regression were applied to analyse patterns in the data. Each model learned relationships between factors like workload, rest/sleep, workplace support, and WLB scores to classify individuals into different WLB levels. After evaluating and comparing their performance based on accuracy and other metrics, the most effective model was selected. The final model was then used to predict WLB levels and support further analysis.

### **Model Selection**

After evaluating multiple machine learning models, the model that achieved the highest accuracy and best overall performance was selected to classify Work-Life Balance (WLB).

## **Evaluation and Validation**

The performance of the models and analysis was evaluated using metrics such as accuracy, confusion matrix, and classification report. These measures helped assess the predictive performance and consistency of the models, ensuring the reliability and effectiveness of the results.

## **Feature Importance Analysis**

Feature importance analysis was performed using the Random Forest model to identify the most influential factors affecting Work-Life Balance (WLB).

## **Recommendation System**

A recommendation system was developed based on the computed Work-Life Balance (WLB) score, which provides personalized suggestions by analysing key factors influencing an individual's work-life balance. The system uses model outputs and score thresholds to generate appropriate recommendations for improving overall balance.

## **Data Description**

The dataset used in this study consists of responses collected from working individuals across various sectors through a structured questionnaire. The dataset contains **549 observations** and multiple variables, including both demographic and work-life balance-related factors. The demographic variables include gender, age group, marital status, family type, occupation and job role, while the work-life balance variables include workload, work beyond office hours, stress level, rest/sleep, personal time, family time, work-life balance perception and workplace support.

The questionnaire was designed using a Likert scale to capture respondents' perceptions. To enable quantitative analysis, responses were converted into numerical values, as shown in the table below. This coding allowed the data to be used for statistical analysis and machine learning model building. To maintain consistency in the direction of measurement, negatively framed variables such as work beyond office hours and stress level were reverse coded so that higher values consistently represent better work-life balance.

For ease of analysis and implementation in statistical and machine learning models, the variables were renamed into simplified and standardized forms. The mapping of original questionnaire items to their respective variable names is presented below.

### **Variable Rename**

| <b>Original Questionnaire Statement</b>                                  | <b>Short Variable Name</b> |
|--------------------------------------------------------------------------|----------------------------|
| My workload is manageable during regular working hours.                  | Workload                   |
| I often have to work beyond official working hours.                      | WorkBeyondTime             |
| I feel not stressed trying to manage work and personal responsibilities. | StressManage               |
| I get enough rest and sleep on working days.                             | RestSleep                  |
| I am able to balance my work and personal life effectively.              | WorkLifeBalance            |
| My job allows me enough time for my family/social life.                  | FamilyTime                 |
| Workplace support.                                                       | WorkplaceSupport           |

### **Likert Scale Coding**

| <b>Response Category</b> | <b>Assigned Values</b> |
|--------------------------|------------------------|
| Strongly Disagree        | 1                      |
| Disagree                 | 2                      |
| Neutral                  | 3                      |
| Agree                    | 4                      |
| Strongly Agree           | 5                      |

## **TERMINOLOGY**

The data is analysed using Jupyter Notebook, MS-Excel, VS code. Statistical tools like Machine learning algorithm.

### Software, Libraries and Packages Used

| <b>Term</b>            | <b>Description</b>                                                                                                                                                                              |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NumPy                  | NumPy is a basic Python library used for numerical computations. It helps in working with arrays, matrices, and performing mathematical operations needed for data analysis and model building. |
| Pandas                 | Pandas is used for handling and analysing data. It makes it easy to clean, organize, and transform raw data into structured formats like Data Frames.                                           |
| Matplotlib             | Matplotlib is used to create simple graphs like bar charts, histograms, and line plots to understand data patterns.                                                                             |
| Seaborn                | Seaborn is built on Matplotlib and is used to create more attractive and informative statistical graphs such as heatmaps and distribution plots.                                                |
| Scikit-learn (sklearn) | Scikit-learn is used for building machine learning models. It provides tools for classification, prediction, and evaluation of models.                                                          |
| SciPy                  | SciPy is used for scientific and statistical computations like hypothesis testing and probability calculations.                                                                                 |
| Stats models           | Stats models is used for statistical modelling and analysis. In this project, it is used for implementing Ordinal Logistic Regression.                                                          |
| Joblib                 | Joblib is used to save and load trained machine learning models so that they can be reused without retraining.                                                                                  |
| Factor Analyzer        | Factor Analyzer is used to perform factor analysis, which helps in reducing multiple variables into fewer important factors.                                                                    |
| Recommendation System  | The recommendation system is developed using model predictions to provide suggestions. It analyses inputs and gives ways to improve work-life balance.                                          |

## **Machine Learning**

Machine learning (ML) is a branch of Artificial Intelligence (AI) that enables systems to learn from data and make predictions or decisions without being explicitly programmed. Instead of writing fixed rules, models are trained using data so that they can identify patterns and relationships. In this study, machine learning is used to predict work-life balance levels based on different factors such as workload, stress and workplace support.

## **Supervised Learning**

Supervised learning is a type of machine learning where the model is trained using labelled data. Each input variable has a corresponding output, which helps the model learn the relationship between them. The trained model can then be used to make predictions on new unseen data.

## **Classification**

Classification is a supervised learning technique where data is categorized into different groups. The model learns from labelled data and assigns new data points to predefined categories. In this project, classification is used to divide work-life balance into categories such as low, moderate and high.

## **Model**

A model is a trained system that learns patterns from data and is used to make predictions. It simplifies real-world problems by focusing on important relationships between variables.

## **CART**

CART (Classification and Regression Tree) is a supervised machine learning algorithm used for both classification and regression tasks. It works like a flowchart, where the data is split into smaller groups based on certain conditions. Each split represents a decision and the final outcome is shown at the leaf nodes. The model selects the best variable for splitting using measures like the Gini Index, which helps in forming more accurate and pure groups. It is simple to understand, easy to interpret and widely used for analysis.

## **CART with Grid Search**

CART with Grid Search refers to an improved version of the Decision Tree model where different parameter combinations are tested to find the best performing model. Grid Search is a technique used to automatically tune hyperparameters such as tree depth, minimum samples split and minimum samples per leaf. It tries multiple combinations and selects the one that gives the highest accuracy. This helps in improving model performance and avoids problems like overfitting or underfitting, making the model more reliable for prediction.

### **Random Forest**

Random Forest is an advanced machine learning algorithm that combines multiple decision trees to improve accuracy and reduce overfitting. It works by creating different subsets of the data and building a decision tree for each subset.

Each tree gives its own prediction and the final result is obtained using majority voting. This makes Random Forest more stable and accurate compared to a single decision tree.

### **Ordinal Logistic Regression**

Ordinal Logistic Regression is used when the dependent variable has ordered categories. Unlike simple logistic regression, which is used for two categories, this model can handle multiple ordered levels.

In this project, it is used to predict work-life balance levels such as low, moderate and high. The model estimates the probability of an observation belonging to each category based on the input variables.

### **Model Evaluation**

Model evaluation is used to measure the performance of machine learning models and to compare different models.

#### **Accuracy:**

Accuracy measures how many predictions made by the model are correct.

#### **Precision:**

Precision shows how many predicted positive cases are actually correct.

#### **Recall:**

Recall shows how many actual positive cases are correctly identified.

#### **F1-Score:**

F1-score provides a balance between precision and recall.

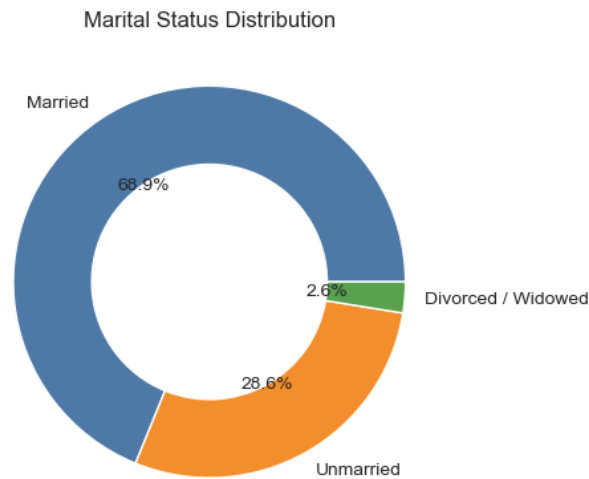
### **Non-Parametric Methods**

Non-parametric methods are statistical techniques that do not depend on any fixed data distribution. They are suitable for data that does not follow normal patterns. Some common non-parametric methods include the Mann-Whitney U test, Kruskal-Wallis test and Dunn's test which are used for comparing groups.



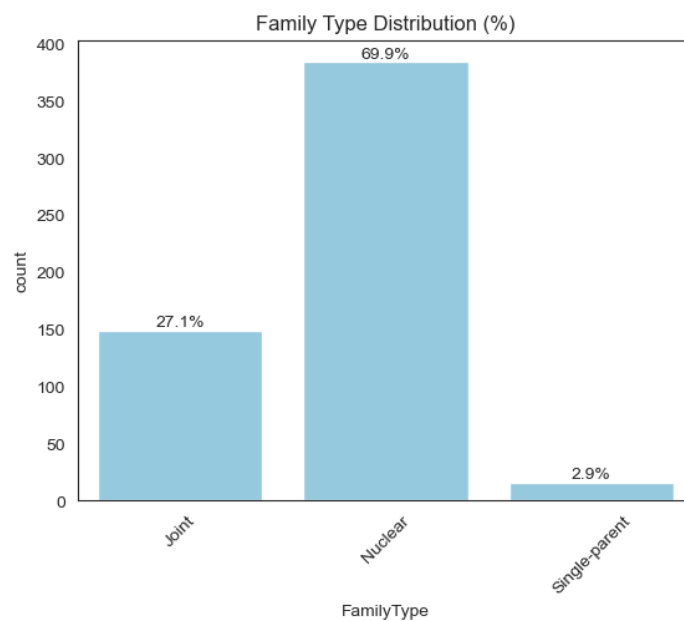
## EXPLORATORY DATA ANALYSIS

### 1) Marital Status Distribution



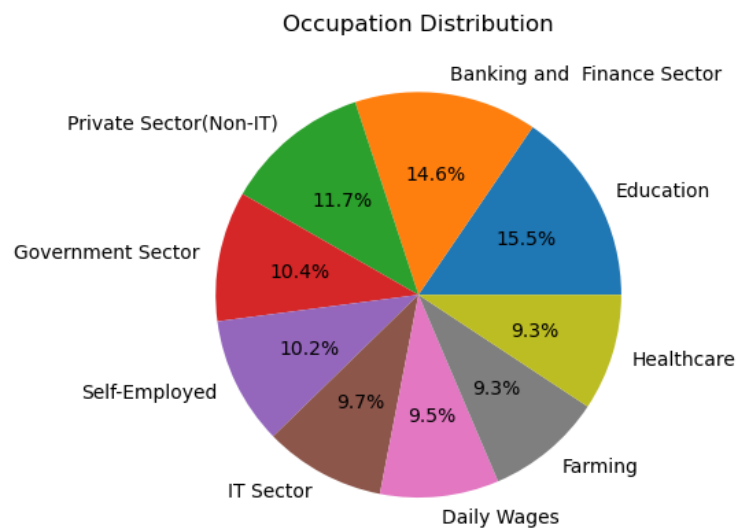
**Interpretation:** - The chart shows that distribution of respondents based on their marital status. It is observed that 68.9% of respondents are married, 28.6% are unmarried and only 2.6% are divorced or widowed.

### 2) Family Type Distribution:



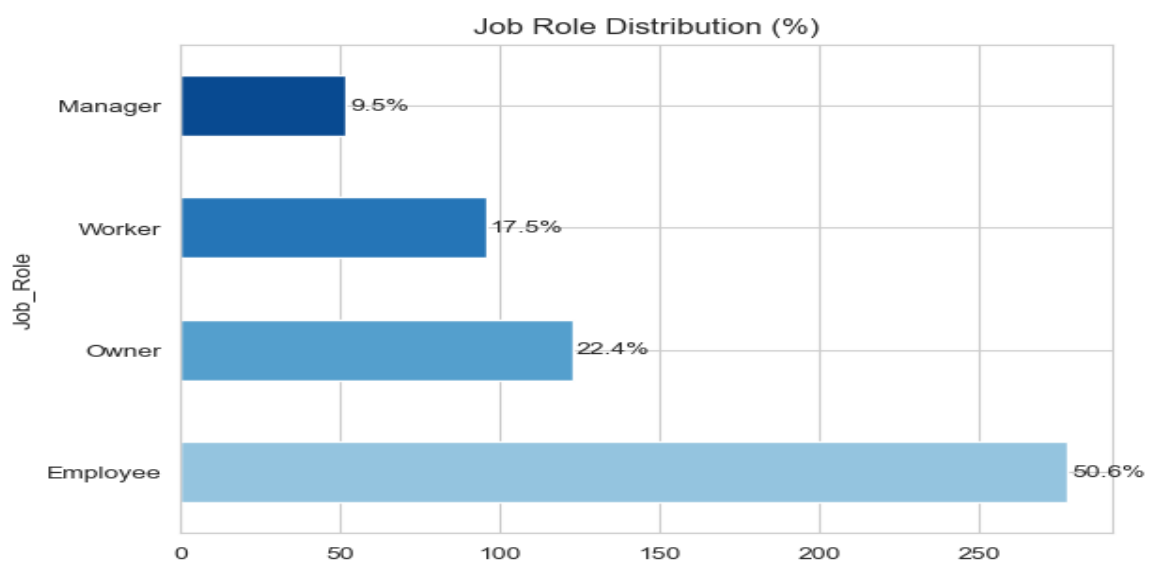
**Interpretation:** - The graph shows that 69.9% respondents belong to nuclear families, 27.1% to joint families and only 2.9% are from single-parent families. This indicated that most participants live in nuclear family.

### 3) Occupation Distribution: -



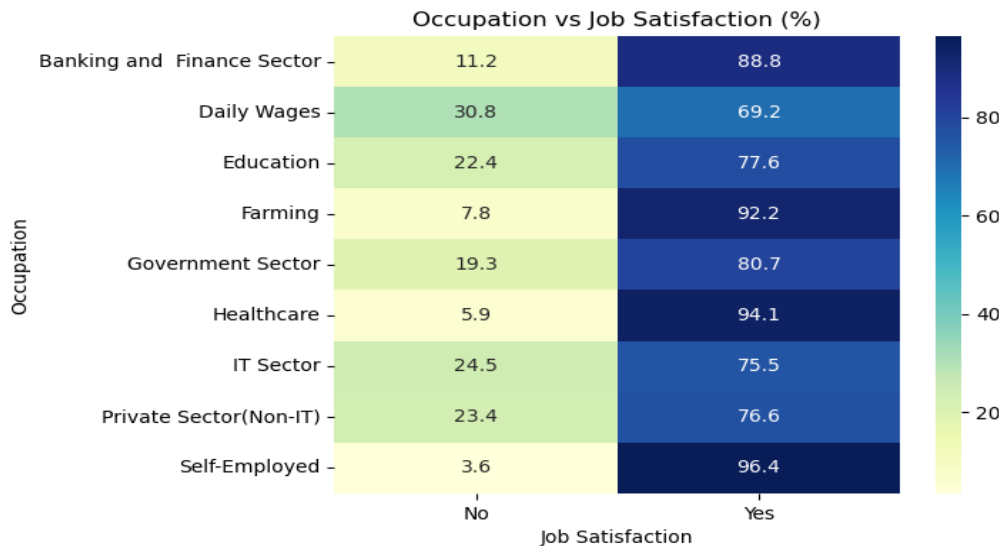
**Interpretation:** -The respondents are distributed across multiple sectors. The highest representation is from Education (15.5%) and Banking and Finance (14.6%), followed by other sectors like private, government, IT etc., each contributing around 9-12%.

### 4) Job Role Distribution:



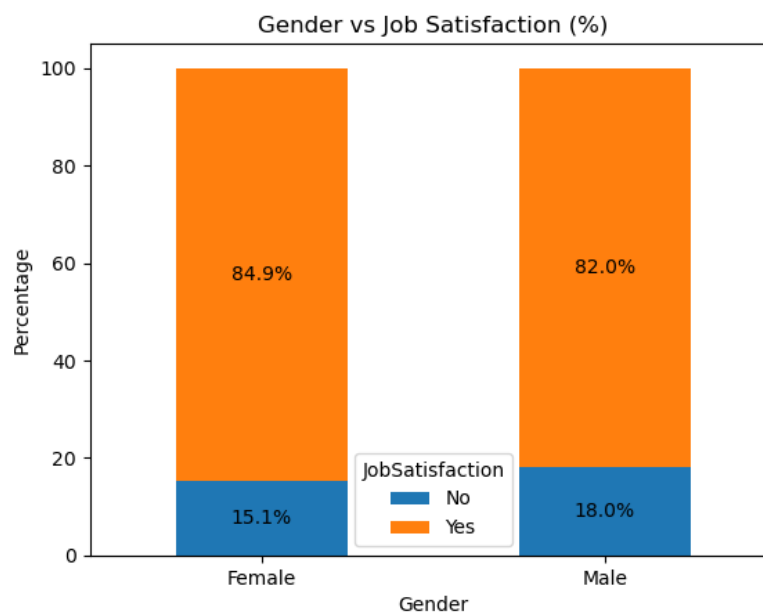
**Interpretation:** - Most respondents are employees (50.6%), followed by owners (22.4%), workers (17.5%) and Managers (9.5%).

### 5) Occupation vs Job Satisfaction:



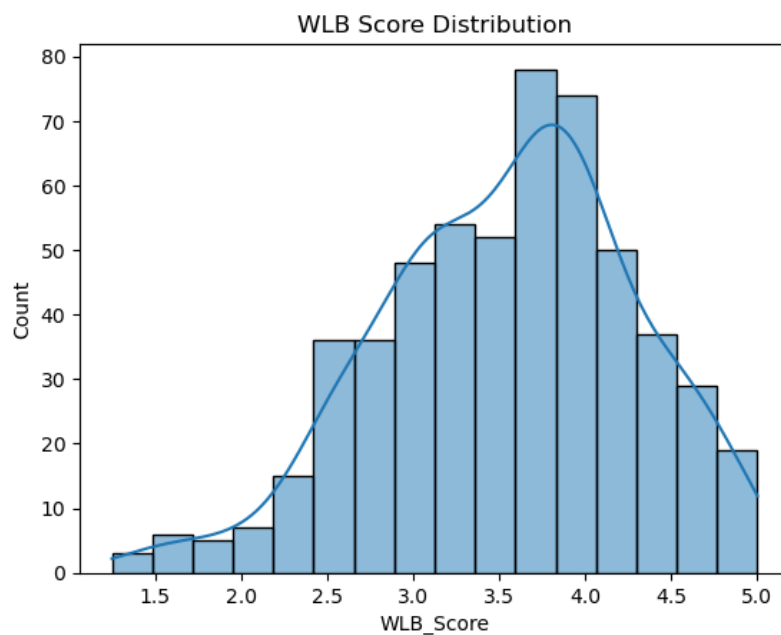
**Interpretation:** - Most occupations show high job satisfaction. The highest job satisfaction is seen in self-employed (96.4%) and healthcare (94.1%), while relatively lower satisfaction is seen in daily wages (69.2%) and IT sector (75.5%).

### 6) Gender vs Job Satisfaction:



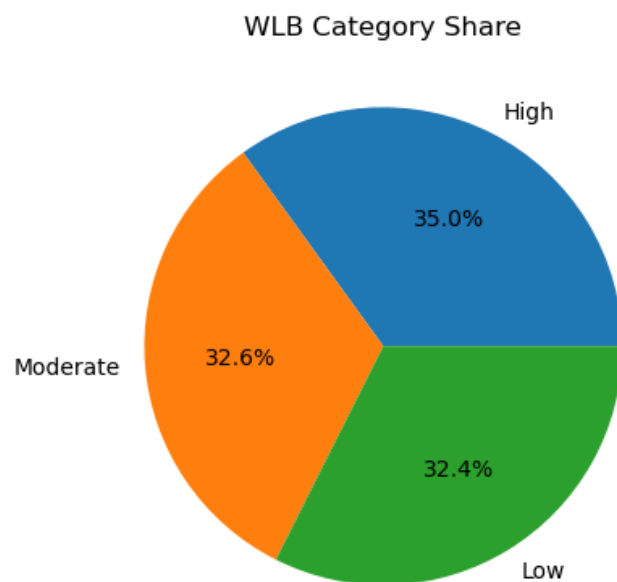
**Interpretation:** - Both genders show approximately same job satisfaction.

### 7) Density Curve (WLB Score Distribution):



**Interpretation:** -The scores are mostly lies between 3 to 4, forming non-normal distribution. Very few respondents have extremely low or high scores.

### 8) WLB Category Distribution:

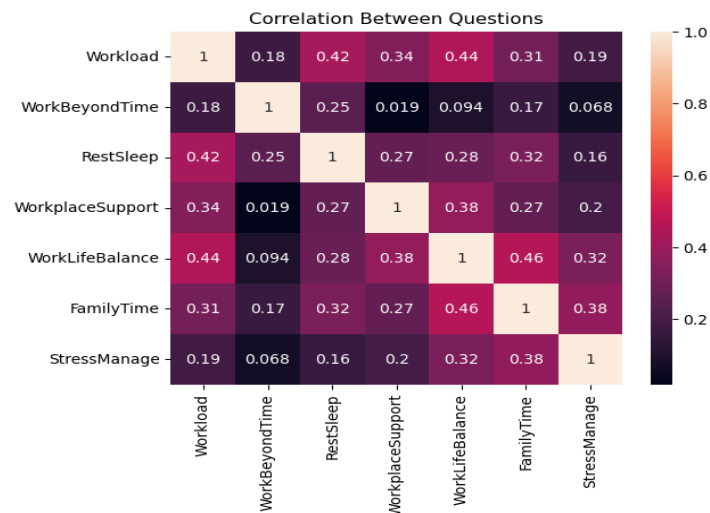


**Interpretation:** -

The distribution is almost equal: (High:35.0%, Moderate:32.6%, Low:32.4%).

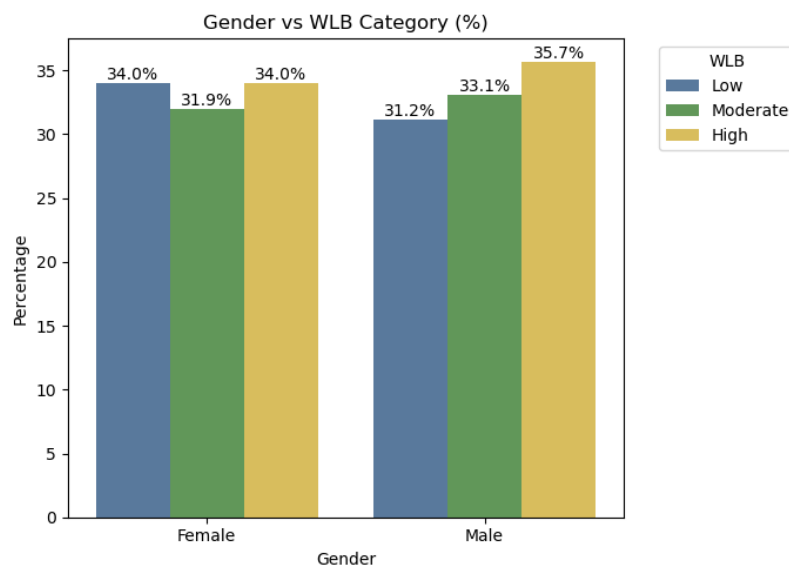
WLB levels are balanced across all categories.

## 9) Correlation Heatmap:



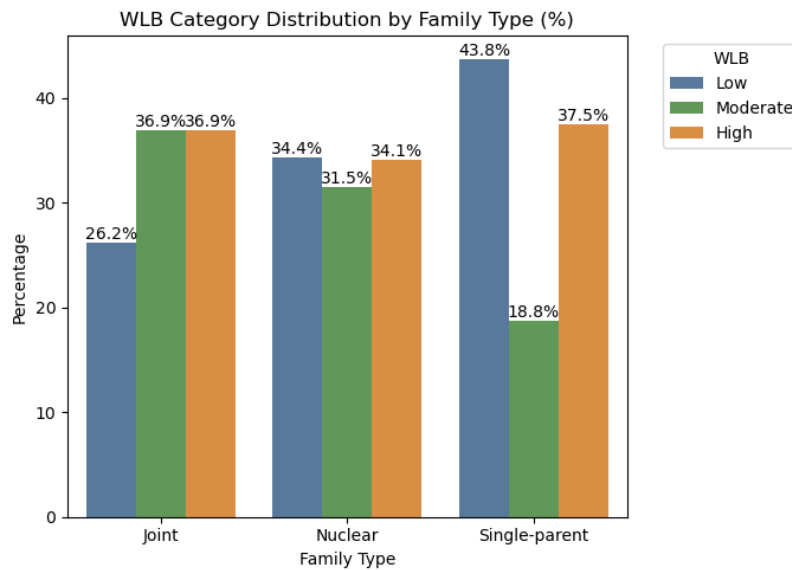
**Interpretation:** - The graph shows how different factors are related to each other. It can be seen that some factors like workload, rest/sleep, family time and workplace support are moderately connected. This means these factors affect each other in some way. Overall, both personal life and work-related factors are linked together and play a role in maintaining balance.

## 10) Gender vs WLB:



**Interpretation:** - Both genders show similar distribution. Males have slightly higher WLB (35.7%) than females (34.0%). WLB nearly equal across genders.

## 11) WLB by Family Type:

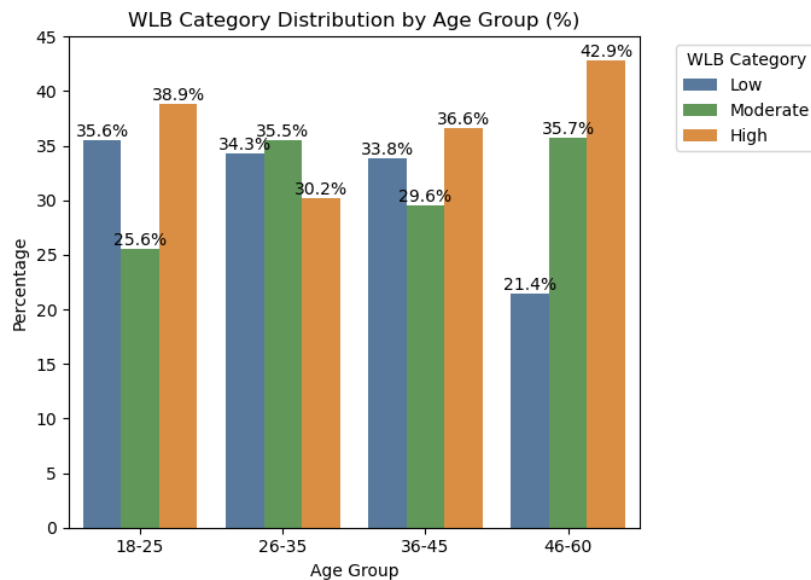


**Interpretation: -**

- Joint families → balanced WLB.
- Nuclear families → moderate distribution.
- Single-Parent → higher low WLB (43.8%).

**Single-Parent families face more imbalance and Family support improves WLB.**

**12) WLB by Age Group**



**Interpretation: -**

Age 46-60 shows highest WLB (42.9%). Younger groups have slightly lower balance.

WLB improves as age and experience increases. It means that older employees manage balance better.

## **STATISTICAL ANALYSIS**

### **Normality**

#### **Shapiro-Wilk Normality Test**

The Shapiro-Wilk test was used to check whether the Work-Life Balance (WLB) score follows a normal distribution or not. This step is important before selecting statistical methods, because many tests assume that the data is normally distributed. By performing this test, we were able to understand the nature of the data and decide whether to use parametric or non-parametric methods for further analysis.

#### **Hypothesis to be tested:**

**H0:** The Work-Life Balance (WLB) score follows a normal distribution.

V/s

**H1:** The Work Life Balance (WLB) score does not follow a normal distribution.

#### **Output:**

Statistic = 0.9863

p-value = 5.495137356518739e-05

#### **Decision Criteria:**

Here p-value is less than significance level of 0.05. Hence, we reject H0.

#### **Conclusion:**

The Work-Life Balance (WLB) score does not follow a normal distribution.

### **Mann-Whitney U Test**

The Mann-Whitney U test is used to compare two groups when the data does not follow a normal distribution. It helps in understanding whether there is a difference between the two groups based on their ranks rather than actual values. This test is useful when the assumptions of parametric tests are not satisfied.

#### **1. Gender-wise Comparison of WLB Scores**

#### **Hypothesis to be tested:**

**H0:** There is no significant difference in Work-Life Balance (WLB) scores between male and female employees.

V/s



**H1:** There is significant difference in Work-Life Balance (WLB) scores between male and female employees.

**Output:**

U Statistic: 38108.0

p-value: 0.526912

**Decision Criteria:**

Here p-value is greater than significance level of 0.05. Hence, we accept H0.

**Conclusion:**

There is no significant difference in Work-Life Balance (WLB) scores between male and female employees.

## **2.Marital Status-wise Comparison of WLB Scores**

**Hypothesis to be tested:**

**H0:** There is no significant difference in Work-Life Balance (WLB) scores between married and unmarried employees.

V/s

**H1:** There is significant difference in Work-Life Balance (WLB) scores between married and unmarried employees.

**Output**

Married: 378

Unmarried: 157

U Statistic: 31171.0

p-value: 0.3292

**Decision Criteria**

Here p-value is greater than significance level of 0.05. Hence, we accept H0.

**Conclusion:**

There is no significant difference in Work-Life Balance (WLB) scores between married and unmarried employees.

### **Kruskal-Wallis Test**

The Kruskal-Wallis test is used when we want to compare more than two groups and the data is not normally distributed. Instead of using actual values, it compares the ranks of the data to check if there is any difference between the groups. This test is useful when normal distribution conditions are not satisfied.

#### **Hypothesis to be tested:**

**H0:** All occupation groups have the same distribution of Work-Life Balance (WLB) scores.

**V/s**

**H1:** At least one occupation group differs in its Work-Life Balance (WLB) score distribution.

#### **Output:**

H Statistic: 49.9664

p-value: 4.147832467362876e-08

#### **Decision Criteria:**

Here p-value is less than significance level of 0.05. Hence, we reject H0.

#### **Conclusion:**

There is significant difference in WLB scores across different occupation groups.

### **Post-hoc Dunn's Test**

The post-hoc Dunn's test is used after the Kruskal-Wallis test when we find that there is a difference among multiple groups. It helps to identify exactly which specific groups are different from each other. This test compares the groups in pairs using ranked data.

#### **Output:**

|                            | <b>Banking and Finance Sector</b> | <b>Daily Wages</b> |
|----------------------------|-----------------------------------|--------------------|
| Banking and Finance Sector | 1.000000                          | 1.000000           |
| Daily Wages                | 1.000000                          | 1.000000           |
| Education                  | 1.000000                          | 1.000000           |
| Farming                    | 1.000000                          | 1.000000           |
| Government Sector          | 0.017979                          | 0.224864           |
| Healthcare                 | 0.352903                          | 1.000000           |
| IT Sector                  | 0.539218                          | 1.000000           |
| Private Sector (Non-IT)    | 0.389821                          | 1.000000           |
| Self-Employed              | 0.323694                          | 0.199595           |

|                            | <b>Education</b> | <b>Farming</b> | <b>Government Sector</b> |
|----------------------------|------------------|----------------|--------------------------|
| Banking and Finance Sector | 1.000000         | 1.000000       | 1.797877e-02             |
| Daily Wages                | 1.000000         | 1.000000       | 2.248642e-01             |
| Education                  | 1.000000         | 1.000000       | 4.977985e-02             |
| Farming                    | 1.000000         | 1.000000       | 7.458994e-02             |
| Government Sector          | 0.049780         | 0.074590       | 1.000000e+00             |
| Healthcare                 | 0.777320         | 0.797752       | 1.000000e+00             |
| IT Sector                  | 1.000000         | 1.000000       | 1.000000e+00             |
| Private Sector (Non-IT)    | 0.892378         | 0.939800       | 1.000000e+00             |
| Self-Employed              | 0.107516         | 0.586965       | 6.639331e-07             |

|                            | <b>Healthcare</b> | <b>IT Sector</b> | <b>Private Sector</b> |
|----------------------------|-------------------|------------------|-----------------------|
| Banking and Finance Sector | 0.352903          | 0.539218         | 0.389821              |
| Daily Wages                | 1.000000          | 1.000000         | 1.000000              |
| Education                  | 0.777320          | 1.000000         | 0.892378              |
| Farming                    | 0.797752          | 1.000000         | 0.939800              |
| Government Sector          | 1.000000          | 1.000000         | 1.000000              |
| Healthcare                 | 1.000000          | 1.000000         | 1.000000              |
| IT Sector                  | 1.000000          | 1.000000         | 1.000000              |
| Private Sector (Non-IT)    | 1.000000          | 1.000000         | 1.000000              |
| Self-Employed              | 0.000076          | 0.000136         | 0.000051              |

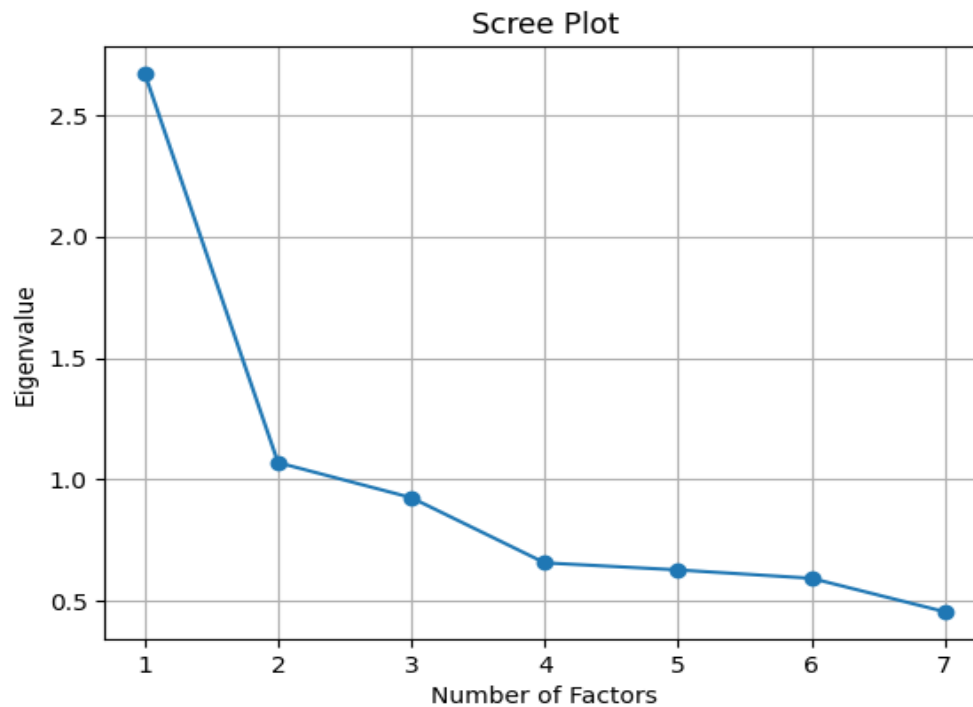
|                            | <b>Self-Employed</b> |
|----------------------------|----------------------|
| Banking and Finance Sector | 3.236938e-01         |
| Daily Wages                | 1.995954e-01         |
| Education                  | 1.075162e-01         |
| Farming                    | 5.869650e-01         |
| Government Sector          | 6.639331e-04         |
| Healthcare                 | 7.600169e-05         |
| IT Sector                  | 1.357793e-04         |
| Private Sector (Non-IT)    | 5.092599e-05         |
| Self-Employed              | 1.0000000e+00        |

**Conclusion:**

Post-hoc Dunn's test with Bonferroni correction shows that significant differences in WLB scores exist between specific occupation groups. The Self-Employed group differs significantly from Government, Healthcare, IT and Private Sector (Non-IT) while the Government sector also differs from Banking & Finance and Education. Overall, WLB varies across occupations rather than being uniform.

## Factor Analysis

### 1. Scree plot:-



A scree plot was constructed to determine the optimal number of factors to retain. The plot shows eigenvalues against the number of factors.

**Interpretation:** The scree plot shows a clear elbow after the second factor, indicating sharp decline in eigenvalues. This suggests that the first two factors capture the majority of the variance in the data while the remaining factors contribute minimally. Therefore, retaining two factors is appropriate.

### 2. Factor Extraction (Kaiser Criterion):

Factor analysis was performed on seven variables related to work-life balance using the Kaiser Criterion.

The eigenvalues obtained were:

- Factor1: 2.67
- Factor2: 1.07
- Remaining factors < 1

**Interpretation:** According to the Kaiser Criterion (Eigenvalue > 1) only two factors were retained for further analysis because they explain a significant portion of the total variance.

### 3.Variance Explained:

The percentage of variance explained by each retained factor is as follows:

- Factor 1: 38.16%
- Factor2: 15.28%

Total Variance Explained = 53.44%.

**Interpretation:** The two extracted factors together explain approximately 53% of the total variance indicating a moderate level of data representation, Factor 1 contributes the most to explaining variability in work-life balance, followed by Factor 2.

### 4.Factor Interpretation using Factor Loadings

| Variables                                                             | Factor 1 | Factor 2 |
|-----------------------------------------------------------------------|----------|----------|
| Workplace support                                                     | 0.445    | 0.213    |
| I am able to balance my work and personal life effectively            | 0.732    | 0.186    |
| My job allows me enough time for my family / social life              | 0.579    | 0.24     |
| I feel not stressed trying to manage work & personal responsibilities | 0.468    | 0.067    |
| My workload is manageable during regular working hours                | 0.436    | 0.451    |
| I often have to work beyond official working hours                    | 0.064    | 0.319    |
| I get enough rest and sleep on working days                           | 0.229    | 0.733    |

Based on factor loadings, variables were grouped into two meaningful dimensions:

#### Factor 1: Work-Life Balance & Satisfaction

##### Variables Included:

- I am able to balance my work and personal life effectively.
- My job allows me enough time for my family/social life.

- Workplace support
- I feel not stressed trying to manage my work & personal responsibilities.

**Interpretation:**

This factor represents employee's overall satisfaction with their work-life balance. It reflects the ability to manage personal and professional responsibilities, availability of support at the workplace and lower stress level.

**Factor 2: Work Pressure**

**Variables Included:**

- My workload is manageable during regular working hours.
- I often have to work beyond official working hours.
- I get enough rest and sleep on working days.

**Interpretation:**

These factors are related to workload pressure, extended working hours and physical/mental tiredness. It indicated the intensity of job demands and their impact on employee's wellbeing. Overall, the factor analysis reveals that WLB can be effectively summarized into two major dimensions. These two factors provide simplified and meaningful structure for understanding employees WLB. The finding suggests that both supportive work environments and manageable workloads play crucial role in achieving health balance.

## Machine Learning Algorithms

### 1. Classification and Regression Tree Model (CART)

The CART model is used to classify individuals into different work-life balance categories based on various input factors. It works by splitting the data into smaller groups using important variables and forms a tree-like structure. Each split is based on conditions that best separate the data into different categories.

#### Model

$$Y=f(X_1, X_2, X_3, \dots, X_k)$$

Where,

$Y$  = dependent variable (WLB category)

$X_1, X_2, \dots, X_k$  = independent variables

#### Output

Accuracy: 0.6636

Classification Report:

|   | Precision | Recall | F1-Score | Support |
|---|-----------|--------|----------|---------|
| 0 | 0.74      | 0.42   | 0.54     | 33      |
| 1 | 0.55      | 0.73   | 0.63     | 37      |
| 2 | 0.76      | 0.80   | 0.78     | 40      |

|              |      |      |      |     |
|--------------|------|------|------|-----|
| accuracy     |      |      | 0.66 | 110 |
| macro avg    | 0.68 | 0.65 | 0.65 | 110 |
| weighted avg | 0.68 | 0.66 | 0.66 | 110 |

Confusion Matrix:

[[14 15 4]

[ 4 27 6]

[ 1 7 32]]

#### Interpretation:

An accuracy of CART model is 66% indicating moderate predictive performance.



## 1.2 CART Model with Grid Search

The CART model with Grid Search is used to improve the performance of the basic decision tree model. In this approach, different combinations of model parameters are tested to find the best possible settings. This helps in selecting the most accurate and reliable model. By tuning parameters like tree depth and split conditions, the model becomes more efficient and avoids overfitting or underfitting.

### Output

Accuracy: 0.718181

Classification Report:

|   | Precision | Recall | F1-Score | Support |
|---|-----------|--------|----------|---------|
| 0 | 0.71      | 0.61   | 0.66     | 33      |
| 1 | 0.59      | 0.65   | 0.62     | 37      |
| 2 | 0.85      | 0.88   | 0.86     | 40      |

|              |      |      |      |     |
|--------------|------|------|------|-----|
| accuracy     |      |      | 0.72 | 110 |
| macro avg    | 0.72 | 0.71 | 0.71 | 110 |
| weighted avg | 0.72 | 0.72 | 0.72 | 110 |

Confusion Matrix

[[20 12 1]

[ 8 24 5]

[ 0 5 35]]

### Interpretation:

After applying hyperparameter tuning, the CART model achieved an improved accuracy of 72% indicating better performance compared to the earlier model.

## 2.Random Forest Model

The Random Forest model is used to improve prediction accuracy by combining multiple decision trees. Instead of relying on a single tree, it creates many trees using different samples of data and combines their results. This makes the model more reliable and reduces errors, especially overfitting.

### Model

$$Y = (1 / N) \times [ T_1(X) + T_2(X) + \dots + T_n(X) ]$$

Where,

Y = Final predicted WLB category

$T_1(X)$ ,  $T_2(X)$ , ...,  $T_n(X)$  = Predictions from each decision tree

N = Total number of trees

### Output

Accuracy: 0.8090

Classification Report:

|   | Precision | Recall | F1-Score | Support |
|---|-----------|--------|----------|---------|
| 0 | 0.78      | 0.88   | 0.83     | 33      |
| 1 | 0.76      | 0.76   | 0.76     | 37      |
| 2 | 0.89      | 0.80   | 0.84     | 40      |

|              |      |      |      |     |
|--------------|------|------|------|-----|
| accuracy     |      |      | 0.81 | 110 |
| macro avg    | 0.81 | 0.81 | 0.81 | 110 |
| weighted avg | 0.81 | 0.81 | 0.81 | 110 |

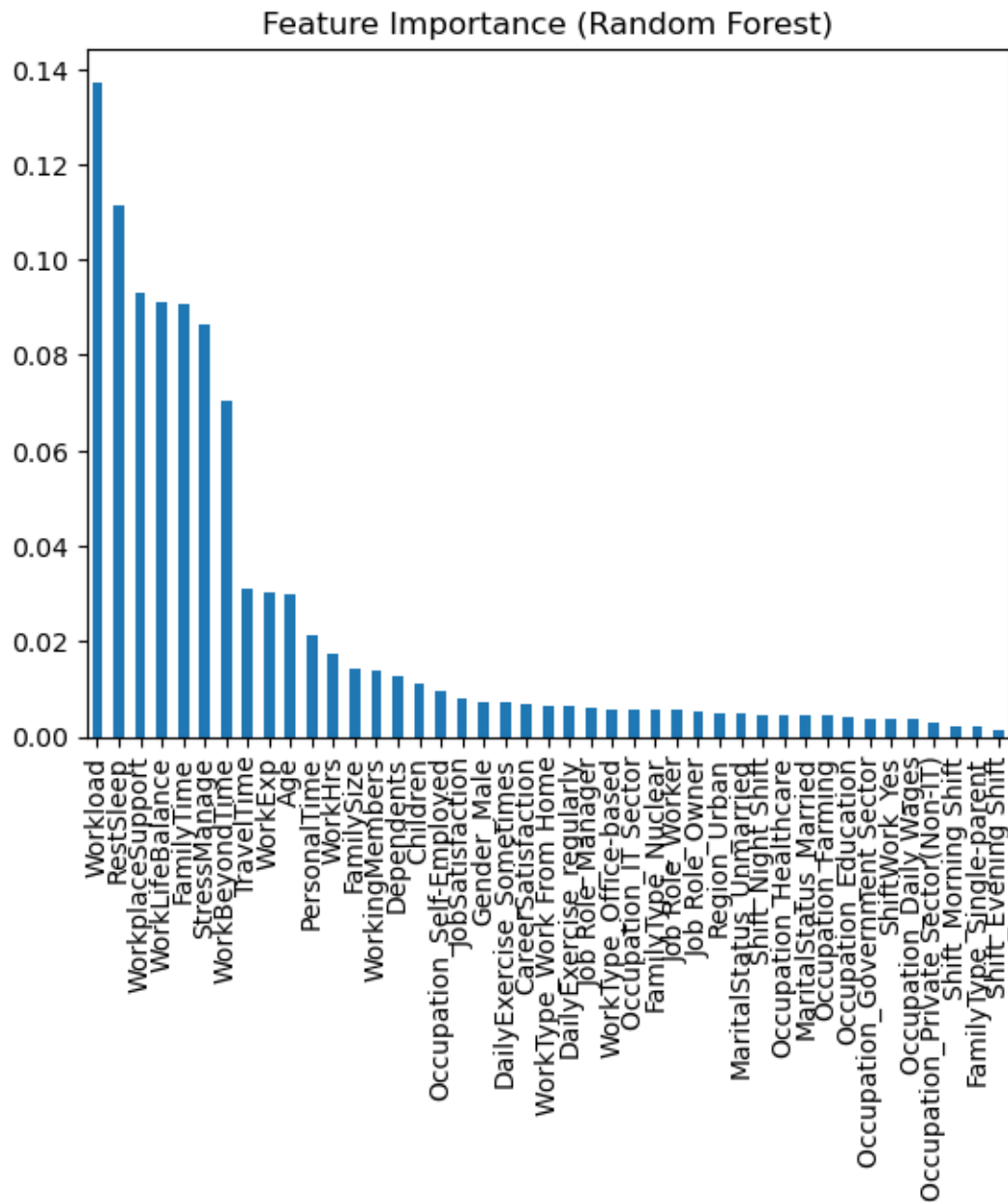
Confusion Matrix:

```
[[29  4  0]
 [ 5 28  4]
 [ 3  5 32]]
```

### Interpretation:

The Random Forest model achieved an accuracy of 81% indicating strong predictive performance.

### **Feature Importance:**



### **Interpretation:**

According to the Feature Importance plot obtained for Random Forest Model, the most important variable affecting the Work-Life Balance is Workload followed by Rest/Sleep, Workplace Support, Family Time and Stress Management. Work related variables including Work Beyond Time and Travel Time are also very important variables whereas age, experience and Personal Time are moderately important. Demographic variables including Occupation Type, Marital Status, Shift Type and Region are least important variables among all the independent variables used in the model. It is evident from the above result that work stress, workplace support and recovery related variables are very important determinants of Work-Life Balance.

### 3. Ordinary Logistic Regression Model

The Ordinal Logistic Regression model is used because the dependent variable (Work-Life Balance) has ordered categories such as low, moderate, and high. This model helps in understanding how different factors influence the probability of an individual falling into these ordered levels. It is suitable for analysing relationships when the outcome variable follows a natural order.

The Ordinal Logistic Regression model was fitted by using demographic and work-related variables such as age, gender, marital status, family type, occupation, job role, work experience, family type and travelling time as independent variables, while the dependent variable was the categorized WLB level.

#### Model:

$$\text{Log}((P(Y>j)/P(Y\leq j))=\alpha_j-(\beta_1X_1+\beta_2X_2+\dots+\beta_kX_k)$$

Where,

$\alpha$  is the intercept,  $\beta_1, \beta_2, \dots, \beta_k$  are coefficients, and  $X_1, X_2, \dots, X_k$  are independent variables.

#### Output

Accuracy: 0.8909

Classification Report:

|   | Precision | Recall | F1-Score | Support |
|---|-----------|--------|----------|---------|
| 0 | 0.88      | 0.88   | 0.88     | 33      |
| 1 | 0.82      | 0.86   | 0.84     | 37      |
| 2 | 0.97      | 0.93   | 0.95     | 40      |

|              |      |      |      |     |
|--------------|------|------|------|-----|
| accuracy     |      |      | 0.89 | 110 |
| macro avg    | 0.89 | 0.89 | 0.89 | 110 |
| weighted avg | 0.89 | 0.89 | 0.89 | 110 |

Confusion Matrix:

```
[[29  4  0]
 [ 4 32  1]
 [ 0  3 37]]
```

**Interpretation:** The model achieved an accuracy of 89% indicating very strong predictive performance.

#### 4. Logistic Regression Model

Logistic Regression is a supervised machine learning model used for classification problems. In this study, it is used to predict job satisfaction of individuals based on different demographic and work-related factors. Since the response variable is binary (satisfied or not satisfied), this model is suitable for analysing the probability of an individual being satisfied with their job.

The model was developed using the following independent variables: age, work experience, travelling time, number of family members, number of dependents, and number of working members in the family. The dependent variable used in the model is job satisfaction.

##### Model:

$$\Pi(X) = 1 / (1 + e^{-(\beta_0 + \beta_1 \text{Age} + \beta_2 \text{WorkExperience} + \beta_3 \text{TravellingTime} + \beta_4 \text{FamilyMembers} + \beta_5 \text{Dependents} + \beta_6 \text{WorkingMembers})})$$

##### Output

Accuracy: 0.6454

##### Classification Report:

|   | Precision | Recall | F1-Score | Support |
|---|-----------|--------|----------|---------|
| 0 | 0.32      | 0.48   | 0.38     | 25      |
| 1 | 0.82      | 0.69   | 0.75     | 85      |

|              |      |      |      |     |
|--------------|------|------|------|-----|
| accuracy     |      |      | 0.65 | 110 |
| macro avg    | 0.57 | 0.59 | 0.57 | 110 |
| weighted avg | 0.70 | 0.65 | 0.67 | 110 |

##### Confusion Matrix:

[[12 13]

[26 59]]

##### Interpretation:

The Logistic regression model achieved an accuracy of 0.6454

#### 4.1 Grid Search Model

Grid Search is used to find the best combination of parameters for the model to improve its performance. Instead of using default values, it tries different values of parameters like C and solver and checks which combination gives the best results. It works by testing multiple models using cross-validation and selecting the one with the highest accuracy. In this study, Grid Search was applied to Logistic Regression to obtain the optimal model for predicting job satisfaction.

##### Output

Accuracy: 0.6909

Classification Report:

|   | Precision | Recall | F1-Score | Support |
|---|-----------|--------|----------|---------|
| 0 | 0.38      | 0.56   | 0.45     | 25      |
| 1 | 0.85      | 0.73   | 0.78     | 85      |

|              |      |      |      |     |
|--------------|------|------|------|-----|
| accuracy     |      |      | 0.69 | 110 |
| macro avg    | 0.61 | 0.64 | 0.62 | 110 |
| weighted avg | 0.74 | 0.69 | 0.71 | 110 |

Confusion Matrix

[[14 11]

[23 62]]

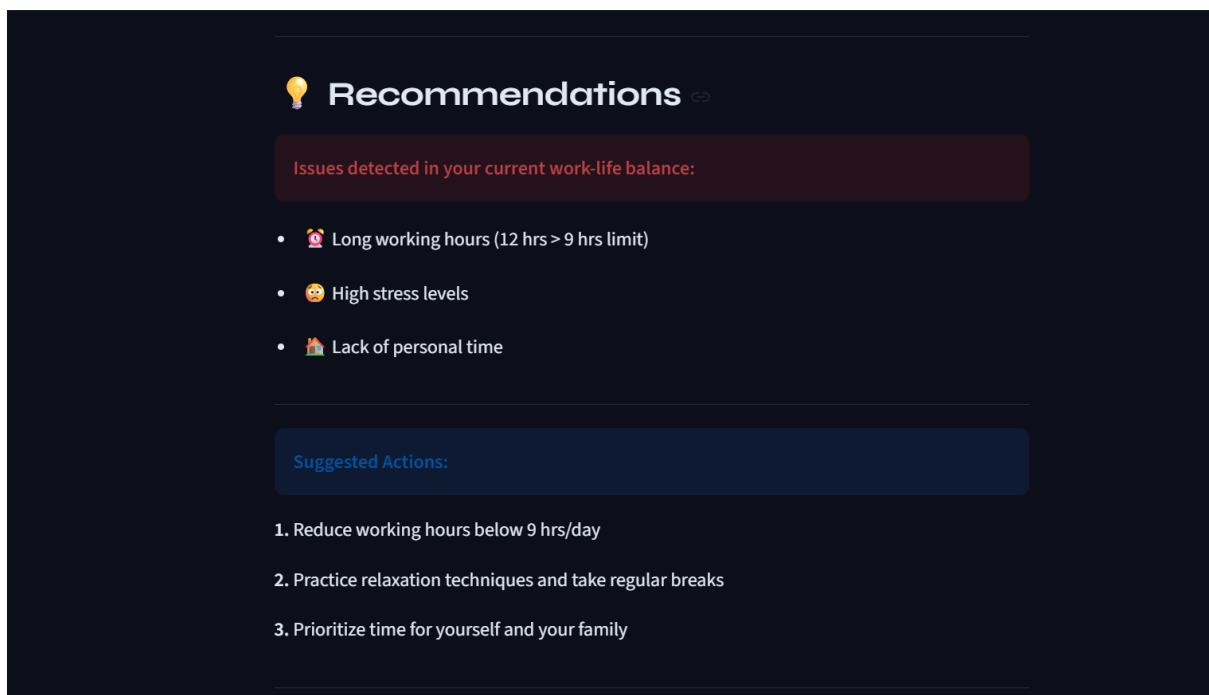
##### Interpretation:

The model after applying Grid Search shows an accuracy of around 69%, indicating a moderate improvement in performance.

### All Models Accuracy

| Model                        | Accuracy | Precision | Recall | F1-Score |
|------------------------------|----------|-----------|--------|----------|
| CART                         | 0.6636   | 0.68      | 0.66   | 0.66     |
| CART Grid Search             | 0.7181   | 0.72      | 0.72   | 0.72     |
| Random Forest                | 0.8090   | 0.81      | 0.81   | 0.81     |
| Ordinary Logistic Regression | 0.8909   | 0.89      | 0.89   | 0.89     |
| Logistic Regression          | 0.70     | 0.65      | 0.67   | 110      |

## Recommendation System





Sector: IT · Limit ≤ 9 hrs · You work 12 hrs · Stress 5/5 · Support 5/5 · Personal 1 hrs/day

 Start Over

### **Interpretation:**

The recommendation system operates by collecting the data from the user in form of input in terms of number of working hours, level of stress, amount of personal time, and level of support at workplace. The system uses a set formula to calculate Work-Life Balance (WLB) score. Upon calculation of WLB score, the level of work-life balance achieved by the user is classified into categories such as low, moderate or high. Then, inputs are analysed in order to find out the source of imbalance. Based on the analysis, suggestions to help increase WLB level are provided. On the whole, this system is an intelligent one since it provides guidance on how WLB can be improved.

## **DISCUSSION**

The results of this study show that work-life balance is affected by many factors, not just one. Both work-related and personal factors play an important role in maintaining balance.

The analysis shows that work-life balance is almost equally distributed among individuals, meaning some people manage it well while others still face difficulties. It was also found that work-life balance is similar across gender and marital status, but it differs across occupations, which means job type has a strong impact.

Factors like family time, workload, and workplace support were found to be very important. People with better support and manageable workload tend to have better work-life balance.

The factor analysis showed that work-life balance can be understood through two main aspects: one related to satisfaction and support, and the other related to work pressure.

Overall, the study shows that improving work-life balance requires both better work conditions and proper personal time management.

## **CONCLUSIONS**

- The study successfully constructed a Work-Life Balance (WLB) score to measure the overall balance of individuals.
- It was found that work-life balance is influenced by key factors such as workload, stress, personal time, and workplace support.
- Factors like family time, manageable workload, and support at the workplace play an important role in maintaining better work-life balance.
- The study identified two main dimensions of work-life balance: one related to satisfaction and support, and the other related to work pressure and job demands.
- The comparison across different groups shows that work-life balance is almost similar across gender;and marital status, but it varies across different occupations.
- The predictive models developed were able to classify individuals into different work-life balance levels effectively.
- The Logistic Regression model shows that job satisfaction is influenced by multiple demographic and work-related factors and can be predicted to a moderate extent. This indicates that job satisfaction depends on a combination of factors rather than a single variable.
- The recommendation system provided useful and personalized suggestions to improve work-life balance based on individual conditions.
- Overall, maintaining a good work-life balance requires both organizational support and effective personal time management, which leads to better well-being and job satisfaction.

## **SUGGESTIONS**

The study shows that improving work-life balance is important for both employees and organizations. Organizations should focus on reducing workload, avoiding long working hours, and creating a supportive work environment. For employees who live far from their workplace, work-from-home opportunities should be provided to reduce travel stress and save time. At the same time, individuals should manage their time properly, take enough rest, and maintain a clear balance between work and personal life. Following these practices can help improve overall well-being, job satisfaction, and productivity.

## **SCOPE AND LIMITATIONS**

### **Scope:**

- Understanding how employees manage their work responsibilities along with their personal and family life.
- Identifying important factors that affect work-life balance, such as workload, working hours, job stress, commuting time, flexibility and organizational support.
- Assessing how work-life balance impacts employee productivity, job satisfaction, mental well-being and overall quality of life.
- Studying the role of employers in providing supportive policies like flexible working hours, work-from-home options, leave facilities and wellness programs.
- Comparing work-life balance across different groups such as age, gender, marital status and work experience to understand variations and specific needs.
- Exploring how technology, hybrid work culture and organizational practices influence an employee's ability to maintain work-life balance.
- Providing useful insights and recommendations for organizations to improve work-life balance policies and reduce employee burnout.

### **Limitations:**

- The study is based on survey data, so responses may include personal bias or inaccurate information.
- The sample size may be limited, so results may not represent all employees or industries.
- Some important factors like personal habits, family situation and detailed company culture are not fully included.
- External factors such as workload changes, company rules or economic conditions are not controlled.
- Rapid changes in technology and work culture may affect the future relevance of the study.

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## APPENDIX

```
#Data import, data cleaning, data preprocessing
import pandas as pd
import numpy as np
# Load dataset
df=pd.read_excel("C:\\Users\\kajal\\Downloads\\WLB_DATA_SET_FINALY_SORTED.xlsx")
print(df.head())
print(df.shape)
df.info()
# Remove extra spaces from column names
df.columns = df.columns.str.strip()
# Replace spaces with underscore
df.columns = df.columns.str.replace(" ", "_")
print(df.columns)
df = df.drop_duplicates()
print(df.isnull().sum())
# fill missing values
df["Workplace_support"] = df["Workplace_support"].fillna("Agree")
df["Do_you_have_any_diagnosed_health_condition?"] = df["Do_you_have_any_diagnosed_health_condition?"].fillna("None")
print(df["Workplace_support"].isnull().sum())
print(df["Do_you_have_any_diagnosed_health_condition?"].isnull().sum())
# select only object (string) columns
str_cols = df.select_dtypes(include="object").columns
# remove extra spaces
df[str_cols] = df[str_cols].apply(lambda x: x.str.strip())
df["Gender"] = df["Gender"].str.capitalize()
# Likert scale coading
likert_scale = {
    "Strongly disagree":1,
    "Disagree":2,
```



```

    "Neutral":3,
    "Agree":4,
    "Strongly agree":5
}
df.replace(likert_scale, inplace=True)
numeric_cols = [
    "Age",
    "Number_of_children",
    "Number_of_family_members",
    "Number_of_dependents(other_than_children)",
    "Number_of_working_members_in_family",
    "Travelling_Time(From_office_to_home)in_min",
    "Work_Experience(years)"
]
for col in numeric_cols:
    df[col] = pd.to_numeric(df[col], errors="coerce")
# Normalikty check of WLB
data = pd.to_numeric(df.iloc[:, -1], errors='coerce').dropna()
# Shapiro test
stat, p = shapiro(data)
print("Statistic =", stat)
print("p-value =", p)
# Histogram
data = pd.to_numeric(df.iloc[:, -1], errors='coerce').dropna()
sns.histplot(data, kde=True)
plt.title("Histogram of WLB Score")
plt.xlabel("WLB Score")
plt.ylabel("Frequency")
plt.show()
#Mann-Whitney test
from scipy.stats import mannwhitneyu
# Label-Encoder of WLB
mapping = {
    "Low": 0,

```

```

    "Moderate": 1,
    "High": 2
}
df['WLB'] = df['WLB'].map(mapping)
# suppose Gender column = Male/Female
group1 = df[df['Gender '] == 'Male']['WLB']
group2 = df[df['Gender '] == 'Female']['WLB']
# perform test
stat, p = mannwhitneyu(group1, group2, alternative='two-sided')
print("U Statistic:", stat)
print("p-value:", p)
# decision
if p < 0.05:
    print("Reject H0 -> Significant difference")
else:
    print("Fail to reject H0 -> No significant difference")
# Mann-Whitney test for marital status
# ensure numeric
df['WLB'] = pd.to_numeric(df['WLB'], errors='coerce')
# groups (STRING MATCH 🔥 )
group1 = df[df['MaritalStatus'] == 'Married']['WLB']
group2 = df[df['MaritalStatus'] == 'Unmarried']['WLB']
# drop NaN
group1 = group1.dropna()
group2 = group2.dropna()
# check size
print("Married:", len(group1))
print("Unmarried:", len(group2))
# test
stat, p = mannwhitneyu(group1, group2, alternative='two-sided')
print("U Statistic:", stat)
print("p-value:", p)
# decision

```

```

if p < 0.05:
    print("Reject H0 -> Significant difference")
else:
    print("Fail to reject H0 -> No significant difference")
# Kruskal wallis Test
from scipy.stats import kruskal
groups = df['Occupation'].unique()
data_groups = [df[df['Occupation'] == g]['WLB'] for g in groups]
stat, p = kruskal(*data_groups)
print("H Statistic:", stat)
print("p-value:", p)
# Post hoc Dunn test
# install (first time only)
!pip install scikit-posthocs
import scikit_posthocs as sp
import pandas as pd
# drop missing values
df_clean = df[['Occupation', 'WLB']].dropna()
# Dunn test with Bonferroni correction
dunn_result = sp.posthoc_dunn(
    df_clean,
    val_col='WLB',
    group_col='Occupation',
    p_adjust='bonferroni'
)
print(dunn_result)
# Factor Analysis
!pip install factor_analyzer
import pandas as pd
from factor_analyzer import FactorAnalyzer
# Load data
df = pd.read_excel("/content/sample_data/7_Questions.xlsx")
print(df.columns)
# Select only required variables

```

```

data = df[[
'My workload is manageable during regular working hours',
'I often have to work beyond official working hours',
'I get enough rest and sleep on working days',
'Workplace support',
'I am able to balance my work and personal life effectively? ',
'My job allows me enough time for my family / social life?',
'I feel not stressed trying to manage work & personal responsibilities.'
]]
# Drop missing values
data = data.dropna()
# Apply Factor Analysis
# Fit FA without fixing number of factors
fa = FactorAnalyzer(rotation=None)
fa.fit(data)
# Total variables
total_vars = len(data.columns)
# Calculate variance %
variance_explained = [(val/total_vars)*100 for val in ev]
# Print result
for i, (eigen, var) in enumerate(zip(ev, variance_explained)):
    print(f"Factor {i+1}: Eigenvalue = {eigen:.2f}, Variance Explained = {var:.2f}%")
# Cumulative variance
import numpy as np
cum_var = np.cumsum(variance_explained)
print("\nCummulative Variance:")
for i, val in enumerate(cum_var):
    print(f"Up to Factor {i+1}: {val:.2f}%")
## Machine Learning Models
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(
X, y, test_size=0.2, random_state=42
)
from sklearn.tree import DecisionTreeClassifier

```

```

dt = DecisionTreeClassifier(
    criterion='gini', # CART uses Gini index
    max_depth=3,
    min_samples_split=5,
    random_state=42
)
dt.fit(X_train, y_train)
y_pred_dt = dt.predict(X_test)
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
print("\nDecision Tree (CART)")
print("Accuracy:", accuracy_score(y_test, y_pred_dt))
print("\nClassification Report:\n")
print(classification_report(y_test, y_pred_dt))
print("\nConfusion Matrix:\n")
print(confusion_matrix(y_test, y_pred_dt))
## CART GridSearch Model
params = {
    'criterion': ['gini'], # CART
    'max_depth': [3, 5, 7],
    'min_samples_split': [2, 5, 10],
    'min_samples_leaf': [1, 5, 10]
}
grid = GridSearchCV(
    estimator=DecisionTreeClassifier(random_state=42),
    param_grid=params,
    cv=5,
    scoring='accuracy',
    n_jobs=-1
)
grid.fit(X_train, y_train)
print("Best Parameters:", grid.best_params_)
best_cart = grid.best_estimator_
y_pred_tuned = best_cart.predict(X_test)
print("\nTuned CART Accuracy:", accuracy_score(y_test, y_pred_tuned))

```

```

print("\nConfusion Matrix:\n", confusion_matrix(y_test, y_pred_tuned))
print("\nClassification Report:\n", classification_report(y_test, y_pred_tuned))
## Random Forest
from sklearn.ensemble import RandomForestClassifier
rf = RandomForestClassifier(
    n_estimators=100,
    max_depth=7,
    random_state=42
)
rf.fit(X_train, y_train)
y_pred_rf = rf.predict(X_test)
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
print("\nRandom Forest")
print("Accuracy:", accuracy_score(y_test, y_pred_rf))
print("\nClassification Report:\n")
print(classification_report(y_test, y_pred_rf))
print("\nConfusion Matrix:\n")
print(confusion_matrix(y_test, y_pred_rf))
## Ordinary Logistic Regression
from statsmodels.miscmodels.ordinal_model import OrderedModel
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
X_train = np.asarray(X_train).astype(float)
X_test = np.asarray(X_test).astype(float)
y_train = np.asarray(y_train).astype(int)
y_test = np.asarray(y_test).astype(int)
model = OrderedModel(y_train, X_train, distr='logit')
result = model.fit(method='bfgs')
pred_probs = result.predict(X_test)
y_pred = np.argmax(pred_probs, axis=1)
print("Accuracy:", accuracy_score(y_test, y_pred))
print("\nConfusion Matrix:\n", confusion_matrix(y_test, y_pred))
print("\nClassification Report:\n", classification_report(y_test, y_pred))

```

```

## Logistic Regression For Job Satisfaction
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

# Load data
df = pd.read_excel("C:\\Users\\kajal\\Desktop\\WLB_Score with_Coadeddata.xlsx");df
print(df['JobSatisfaction'].unique())
print(df['JobSatisfaction'].value_counts())
print(df.columns)

# Step 2: Select variables (X)
X = df[['Age',
        'WorkExp',
        'TravelTime',
        'FamilySize',
        'Dependents',
        'WorkingMembers']]
y = df['JobSatisfaction']

# Step 3: Train-Test Split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Step 4: Model
model = LogisticRegression(max_iter=1000, class_weight='balanced')
model.fit(X_train, y_train)

# Step 5: Prediction
y_pred = model.predict(X_test)

# Step 6: Evaluation
print("Accuracy:", accuracy_score(y_test, y_pred))
print("\nConfusion Matrix:\n", confusion_matrix(y_test, y_pred))
print("\nClassification Report:\n", classification_report(y_test, y_pred))

from sklearn.model_selection import GridSearchCV
from sklearn.linear_model import LogisticRegression

# Step 1: Define parameters
params = {
    'C': [0.01, 0.1, 1, 10],

```

```

'solver': ['liblinear', 'lbfgs']
}
# Step 2: Grid Search
grid = GridSearchCV(
    LogisticRegression(class_weight='balanced', max_iter=1000),
    param_grid=params,
    cv=5,
    scoring='accuracy'
)
# Step 3: Fit grid search
grid.fit(X_train, y_train)
# Step 4: Best model directly use kara
best_model = grid.best_estimator_
# Step 5: Prediction
y_pred = best_model.predict(X_test)
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
print("Accuracy:", accuracy_score(y_test, y_pred))
print(confusion_matrix(y_test, y_pred))
print(classification_report(y_test, y_pred))
# Recommendation System
# App.py
import streamlit as st
import pandas as pd
st.set_page_config(
    page_title="Work Life Balance · Work-Life Score",
    page_icon="⚖️",
    layout="centered"
)
st.markdown("""
<style>
@import url('https://fonts.googleapis.com/css2?family=Syne:wght@400;600;700;800&family=DM+Sans:wght@300;400;500&display=swap');

```



```

:root {
  --bg: #0c0f1a;
  --surface: #141827;
  --border: #252b42;
  --accent: #6ee7b7;
  --accent2: #38bdf8;
  --danger: #f87171;
  --warn: #fbbf24;
  --text: #e2e8f0;
  --muted: #64748b;
}

html, body, [data-testid="stAppViewContainer"] {
  background: var(--bg) !important;
  color: var(--text) !important;
  font-family: 'DM Sans', sans-serif !important;
}

[data-testid="stHeader"], [data-testid="stToolbar"],
[data-testid="stSidebar"], footer, #MainMenu { display: none !important; }

[data-testid="stMainBlockContainer"] {
  max-width: 680px !important;
  margin: 0 auto !important;
  padding: 2rem 1.5rem 4rem !important;
}

.wlb-hero { text-align:center; padding:2.5rem 0 1.5rem; margin-bottom:1.5rem; }
.wlb-hero .logo { font-size:2.8rem; line-height:1; margin-bottom:0.5rem; }
.wlb-hero h1 {
  font-family:'Syne',sans-serif !important;
  font-size:2.2rem !important; font-weight:800 !important;
  color:var(--text) !important; letter-spacing:-1px; margin:0 !important;
}

.wlb-hero h1 span { color:var(--accent); }
.wlb-hero p { color:var(--muted); font-size:0.95rem; margin-top:0.4rem; }

```

```

.progress-wrap { background:var(--border); border-radius:99px; height:6px; overflow:hidden;
margin-bottom:0.3rem; }
.progress-fill { height:100%; border-radius:99px; background:linear-gradient(90deg,var(--ac-
cent),var(--accent2)); }
.progress-label { font-size:0.75rem; color:var(--muted); text-align:right; margin-bot-
tom:1.2rem; }
.hint { font-size:0.83rem; color:var(--muted); margin-top:0.35rem; margin-bottom:0.5rem; }
div[data-testid="stButton"] > button {
    background:linear-gradient(135deg,var(--accent),var(--accent2)) !important;
    color:#0c0f1a !important;
    font-family:'Syne',sans-serif !important; font-weight:700 !important;
    font-size:0.95rem !important; border:none !important;
    border-radius:10px !important; padding:0.6rem 2rem !important;
    width:100% !important; margin-top:1rem !important;
}
div[data-testid="stButton"] > button:hover { opacity:0.85 !important; transform:translateY(-
1px) !important; }
div[data-baseweb="select"] > div,
div[data-baseweb="base-input"] > input,
div[data-testid="stNumberInput"] input {
    background:#1a2035 !important; border:1px solid var(--border) !important;
    border-radius:10px !important; color:var(--text) !important;
    font-family:'DM Sans',sans-serif !important;
}
div[data-baseweb="select"] svg { color:var(--accent) !important; }
div[data-testid="stSlider"] > div > div > div { background:var(--accent) !important; }
div[data-testid="stSlider"] > div > div > div > div {
    background:var(--accent2) !important;
    border:2px solid var(--accent) !important;
    box-shadow:0 0 10px var(--accent) !important;
}
label, [data-testid="stWidgetLabel"] p { color:var(--muted) !important; font-size:0.82rem
!important; }

```

```

[data-testid="stMetricValue"] {
  font-family:'Syne',sans-serif !important;
  font-size:1.6rem !important; font-weight:800 !important;
}
[data-testid="stMetric"] {
  background:#1a2035 !important;
  border:1px solid var(--border) !important;
  border-radius:12px !important;
  padding:1rem !important;
}
[data-testid="stAlert"] { border-radius:12px !important; font-family:'DM Sans',sans-serif !important; }
hr { border-color:var(--border) !important; margin:1.2rem 0 !important; }
h2, h3 { font-family:'Syne',sans-serif !important; font-weight:700 !important; color:var(--text) !important; }
</style>
""", unsafe_allow_html=True)
# — Data —
sector_limits = {
  "IT": 9, "Private": 9, "Government": 8, "Banking": 8,
  "Healthcare": 10, "Education": 8, "Farming": 11,
  "Daily Wage": 10, "Self-employed": 10
}
SECTOR_ICONS = {
  "IT": "💻", "Private": "🏢", "Government": "🏛️", "Banking": "🏦",
  "Healthcare": "🏥", "Education": "📖", "Farming": "🌾",
  "Daily Wage": "🔨", "Self-employed": "👤"
}
STRESS_LABELS = ["", "😌 Very Calm", "😐 Manageable", "😬 Moderate", "😫 High Stress", "🤯 Burnout"]
SUPPORT_LABELS = ["", "😞 Very Poor", "😓 Below Average", "😐 Average", "😄 Good", "😁 Excellent"]

```

```

# — Session —
if "step" not in st.session_state:
    st.session_state.step = 0
    st.session_state.data = {}
TOTAL_STEPS = 6
# — Recommendation logic —
def smart_recommendation(d):
    issues, suggestions = [], []
    limit = sector_limits.get(d["sector"], 9)
    if d["work"] > limit:
        issues.append(f"🕒 Long working hours ({d['work']} hrs > {limit} hrs limit)")
        suggestions.append(f"Reduce working hours below {limit} hrs/day")
    if d["stress"] >= 4:
        issues.append("😓 High stress levels")
        suggestions.append("Practice relaxation techniques and take regular breaks")
    if d["travel"] > 2:
        issues.append("🚗 Very long commute")
        suggestions.append("Consider relocation or remote work options")
    elif d["travel"] > 1:
        issues.append("🚗 Moderate commute time")
        suggestions.append("Try to reduce commute time where possible")
    if d["support"] <= 2:
        issues.append("🤝 Low workplace support")
        suggestions.append("Communicate openly with your team or management")
    if d["personal"] < 2:
        issues.append("🏠 Lack of personal time")
        suggestions.append("Prioritize time for yourself and your family")
    return issues, suggestions
# — Hero —
st.markdown("""
<div class="wlb-hero">
    <div class="logo">🏴‍☠️</div>

```

```

    <h1>Work Life <span>Balance</span></h1>
    <p>Your personal Work-Life Balance Analyzer</p>
</div>

"""', unsafe_allow_html=True)

# — Progress bar —

step = st.session_state.step
if step < TOTAL_STEPS:
    pct = int((step / TOTAL_STEPS) * 100)
    st.markdown(f"""
        <div class="progress-wrap"><div class="progress-fill"
style="width: {pct}%"></div></div>
        <div class="progress-label">Step {step + 1} of {TOTAL_STEPS}</div>
    """, unsafe_allow_html=True)

# STEP 0 — Sector
if step == 0:
    st.subheader("👋 Which sector do you work in?")
    sector = st.selectbox("", list(sector_limits.keys()))
    icon = SECTOR_ICONS.get(sector, "")
    st.markdown(f'<div class="hint">{icon} {sector} &nbsp;·&nbsp; Recommended ≤ {sec-
tor_limits[sector]} hrs/day</div>', unsafe_allow_html=True)
    if st.button("Continue →"):
        st.session_state.data["sector"] = sector
        st.session_state.step = 1
        st.rerun()

# STEP 1 — Work hours
elif step == 1:
    st.subheader("🕒 How many hours do you work daily?")
    work = st.slider("", 1, 15, 8)
    limit = sector_limits[st.session_state.data["sector"]]
    if work <= limit:
        st.success(f"✅ {work} hrs — within your sector limit ({limit} hrs)")
    else:
        st.error(f"❌ {work} hrs — exceeds sector limit of {limit} hrs")

```

```

if st.button("Continue →"):
    st.session_state.data["work"] = work
    st.session_state.step = 2
    st.rerun()

# STEP 2 — Stress
elif step == 2:
    st.subheader("😓 Stress level at work?")
    st.markdown('<div class="hint">1 = Very calm &nbsp;·&nbsp; 5 = Burnout</div>', unsafe_allow_html=True)
    stress = st.slider("", 1, 5, 3)
    if stress <= 2:
        st.success(STRESS_LABELS[stress])
    elif stress == 3:
        st.info(STRESS_LABELS[stress])
    else:
        st.warning(STRESS_LABELS[stress])
    if st.button("Continue →"):
        st.session_state.data["stress"] = stress
        st.session_state.step = 3
        st.rerun()

# STEP 3 — Travel
elif step == 3:
    st.subheader("🚗 Daily commute time (one way, in minutes)")
    travel_min = st.number_input("", min_value=0, max_value=300, step=5, value=30)
    hrs = travel_min / 60
    if hrs <= 0.5:
        st.success(f"✅ Short commute — {travel_min} min/day")
    elif hrs <= 1:
        st.info(f"⚠️ Moderate commute — {travel_min} min/day")
    else:
        st.warning(f"⚠️ Long commute ({travel_min} min/day) — impacts wellness")
    if st.button("Continue →"):
        st.session_state.data["travel"] = hrs

```

```
st.session_state.data["travel_min"] = travel_min
st.session_state.step = 4
st.rerun()

# STEP 4 — Support
elif step == 4:
    st.subheader("👉 How supportive is your workplace?")
    st.markdown('<div class="hint">1 = Poor &nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&~>1 = Poor &nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&~>', unsafe_al-
low_html=True)
    support = st.slider("", 1, 5, 3)
    if support >= 4:
        st.success(SUPPORT_LABELS[support])
    elif support == 3:
        st.info(SUPPORT_LABELS[support])
    else:
        st.warning(SUPPORT_LABELS[support])
    if st.button("Continue →"):
        st.session_state.data["support"] = support
        st.session_state.step = 5
        st.rerun()

# STEP 5 — Personal time
elif step == 5:
    st.subheader("🏠 Personal time available daily (hours)")
    personal = st.slider("", 0, 10, 3)
    if personal < 2:
        st.error(f"🔴 Only {personal} hrs — too little personal time")
    elif personal < 3:
        st.warning(f"⚠️ {personal} hrs — slightly low personal time")
    else:
        st.success(f"✅ {personal} hrs — healthy personal time")
    if st.button("🔍 Analyze My Balance"):
        st.session_state.data["personal"] = personal
        st.session_state.step = 6
        st.rerun()
```

```
# STEP 6 — RESULTS
```

```
elif step == 6:
```

```
    d = st.session_state.data
```

```
    limit = sector_limits.get(d["sector"], 9)
```

```
    work_factor = d["work"] / limit
```

```
    score = (0.4 * work_factor * 10 + 0.8 * d["travel"] + 0.6 * d["stress"]) - \
```

```
        (0.7 * d["support"] + 0.5 * d["personal"])
```

```
    score = round(score, 2)
```

```
    if score < 4:
```

```
        category, cat_icon = "Good", "✅ "
```

```
    elif score < 8:
```

```
        category, cat_icon = "Moderate", "⚠️ "
```

```
    else:
```

```
        category, cat_icon = "Poor", "❌ "
```

```
    burnout_risk = d["work"] > limit and d["stress"] >= 4
```

```
    issues, suggestions = smart_recommendation(d)
```

```
    if category == "Good":
```

```
        st.success(f'#### {cat_icon} {category} Work-Life Balance \nWLB Score: \n\n{score}')
```

```
    elif category == "Moderate":
```

```
        st.warning(f'#### {cat_icon} {category} Work-Life Balance \nWLB Score: \n\n{score}')
```

```
    else:
```

```
        st.error(f'#### {cat_icon} {category} Work-Life Balance \nWLB Score: \n\n{score}')
```

```
    st.divider()
```



```

icon = SECTOR_ICONS.get(d["sector"], "")
c1, c2, c3, c4 = st.columns(4)
c1.metric("🏠 WLB Score", score)
c2.metric("🔥 Burnout", "High 🟡" if burnout_risk else "Normal 🟢")
c3.metric("🏢 Sector", f"{icon} {d['sector']}")
c4.metric("🚗 Commute", f"{d.get('travel_min', round(d['travel']*60))} min")

st.divider()

st.subheader("💡 Recommendations")

if not issues:
    st.success("🎉 You have a **healthy work-life balance**. Keep it up!")
else:
    st.error("**Issues detected in your current work-life balance:**")
    for issue in issues:
        st.markdown(f"- {issue}")
    st.divider()
    st.info("**Suggested Actions:**")
    for i, s in enumerate(suggestions[:3], 1):
        st.markdown(f"**{i}. {s}")

st.divider()
st.caption(
    f"Sector: **{d['sector']}** · Limit ≤ **{limit} hrs** · "
    f"You work **{d['work']} hrs** · Stress **{d['stress']}/5** · "
    f"Support **{d['support']}/5** · Personal **{d['personal']} hrs/day**"
)

df_out = pd.DataFrame([{"d": d, "Score": score, "Category": category}])
df_out.to_csv("wlb_data.csv", mode="a", header=False, index=False)

st.divider()

```

```

if st.button("🔄 Start Over"):
    st.session_state.step = 0
    st.session_state.data = {}
    st.rerun()

# Model.py
import pandas as pd
from sklearn.ensemble import RandomForestClassifier
df = pd.read_csv("wlb_data.csv")
X = df[["work", "stress", "travel", "support", "personal"]]
y = df["Category"]
model = RandomForestClassifier()
model.fit(X,y)
print("Model trained successfully")

#project.py
{
    "folders": [
        {
            "path": "."
        }
    ],
    "settings": {}
}

#requirement
streamlit
pandas
scikit-learn
matplotlib

```